

HENRY PERDA

Duxbury, MA · perda.h@northeastern.edu · (781) 354-9641 · [LinkedIn](#)

EDUCATION

Northeastern University — Boston, MA

GPA 3.5/4.0

B.S. Mechanical Engineering, Minor in Aerospace Engineering

Expected May 2029

- Relevant coursework: Mechanics of Materials, Thermodynamics, Calculus 3, Statics & Vector Mechanics, Cornerstone of Engineering, Programming with MATLAB & Python
- Certification: Certified SOLIDWORKS Associate (CSWA)

TECHNICAL SKILLS

- CAD & Design: SolidWorks (CSWA), AutoCAD, Fusion, Fusion CAM, GD&T, assembly modeling
- Simulation & Analysis: ANSYS Mechanical (FEA), ANSYS Fluent (CFD), SolidWorks Simulation (FEA & Flow)
- Programming: MATLAB, Python, MicroPython
- Fabrication: Metalworking, machining, 3D printing, SLA 3D printing

ENGINEERING PROJECTS

Mortar-Launched Deployable-Wing UAV Drone - Co-Lead

January 2026 – August 2026

- Co-led design of a CO₂ pneumatic-mortar-launched UAV with mechanically deployable wings, capable of reaching a 1 m² target 1.5 km away; owned wing deployment, control surfaces, FEA/CFD simulation, fabrication, and testing
- Ran ANSYS structural analysis on 3D-printed components and built a combined lift/drag/launch-ballistics model to predict deployment altitude, trajectory, and a 7:1 glide ratio
- Engineered the launcher's pressurization tank, expansion coupler, and launch plate to enable reliable mortar-based deployment
- Executing a low-altitude test program, collecting flight data to inform design revisions as well as to predict full altitude test behavior

Baja SAE Northeastern - Drivetrain Engineer

January 2026 – Present

- Lightweighting the drivetrain by redesigning the driveshaft and single U-joints for 400+ ft-lbs of torque, and reworking engine-mount geometry to eliminate the engine subframe
- Designed jigs for the manufacturing process, including jigs for the CNC lathe chuck as well as for welding and assembly
- Manufactured car parts on a manual lathe and ran self-written CAM programs on the CNC mill for the rear gearbox cover
- Participated in team meetings and design reviews, critiquing designs and suggesting improvements to the overall car

Aero NU Liquid Rocket Team - Structures engineer

June 2026 – Present

- Designing and testing a quick-disconnect coupling for the launchpad propellant line, enabling rapid and safe separation during fueling operations
- Developing rocket quick disconnect parts, performing tolerance stack-ups and material selection to meet fabrication constraints
- Collaborated in weekly team meetings and design reviews to iterate on subsystem designs and resolve engineering trade-offs

Aero NU Airbrakes Team - Member

January 2026 – July 2026

- Developed components for an L-2 high-powered rocket containing an airbrakes module designed to target specific altitudes
- Designed the e-bay sled, assembly jigs, and servo housing, optimizing for low weight and structural integrity under flight loads
- Co-designed the Airbrake module, sizing flap surface area to control apogee and reinforcing the structure to withstand shear forces from aerodynamic loading

WORK EXPERIENCE

Manager - Manomet Lobster Pound

August 2026

- Handle cash management, product ordering, and customer hospitality to maintain smooth service during peak summer demand

Island Creek Oysters

Assistant Manager

June 2026 – August 2026

Kitchen Staff

June 2022 – June 2026

- Promoted to Assistant Manager after joining as kitchen staff in 2022; oversee shift operations and supervise team members
- Manage scheduling, opening/closing procedures, and customer service escalations

Notetaker - Northeastern Disability Access Services (DAS)

Summer 2026

- Took detailed lecture notes and submitted them to support students unable to attend class due to disability

LEADERSHIP & ACTIVITIES

- **ASME** - Participate in college-wide events for mechanical engineering students, including team-building exercises
- **Roxbury Robotics** - Meet weekly with middle schoolers in an afterschool program, teaching them to build and program small LEGO robots for competition